



AI-Powered Collections Strategy



Leveraging Agentic AI for Scalable, Fair,
and Effective Debt Management at
Geldium



How the System Works (1/2): From Data to Action

1 Inputs

- Repayment history (Month_1-6, missed payments)
- Credit utilization, debt-to-income
- Employment status, income, tenure
- Card type; IDs pseudonymised



2 Decision Logic

- Data checks and cleaning
- Logistic regression scores delinquency risk
- Business rules add policy limits
- Output: risk tier plus top reasons



3 Actions

- Low: automated reminder
- Medium: personalised SMS/email and payment-plan offer
- High: priority outreach and hardship offer, agent-reviewed



4 Learning Loop

- Track outcomes: on-time payment, plan take-up
- Compare with control group
- Retrain and recalibrate quarterly
- Review fairness and drift

How the System Works (2/2): Decision Logic by Risk Tier

Risk tier	Trigger (thresholds set after validation)	Action	Handled by
Low	Low model score; no recent late payments	Automated payment reminder	System
Medium	Mid score, or utilization of 50% or more with recent late payments	Personalised SMS/email plus flexible payment-plan offer	System, with monitoring
High	High score, or unemployed/Business-card customer with utilization of 50% or more	Priority outreach and tailored hardship plan	Collections agent approves

Learning loop: every contact is logged with its result. Outcomes are compared against a control group, then used to retrain the model and adjust rules each quarter.

Role of Agentic AI: Autonomous vs Human Oversight

Autonomous (AI acts alone)	Human Oversight (agent decides)
Scoring customers and assigning risk tiers daily	Approving hardship plans and payment deferrals
Sending reminders by SMS/email at the best time	Contacting vulnerable or disputed accounts
Choosing message and channel from past results	Any decision that affects credit terms or fees
Monitoring outcomes, drift and fairness alerts	Reviewing bias findings; approving model updates

Principle: AI handles routine, low-impact steps at scale; people own high-impact decisions and can override any AI action.

Responsible AI Guardrails

- **Fairness:** exclude Location and Age from scoring; test outcomes by group (disparate impact ratio of 0.8 or higher, recall gaps) before launch and monthly.
- **Explainability:** every flag shows its top reasons in plain language (logistic regression coefficients); customers can be told why they were contacted.
- **Compliance:** align with ECOA, fair debt collection rules and GDPR/local privacy law; keep audit logs; collect only necessary data.
- **Human oversight:** review step for high-impact decisions, easy override, and support-first actions with no automatic penalties.

Expected Business Impact

Business KPIs

- 30-day delinquency rate: target 10% relative reduction vs control (pilot)
- Higher cure and repayment rates
- Lower cost per collected dollar
- Fewer agent hours on routine contacts
- Model AUC and recall, fairness ratio of 0.8 or higher

Customer Outcomes

- Earlier, supportive contact before accounts fall behind
- Clear reasons and flexible payment options
- Consistent, fair treatment across groups
- Fewer unnecessary or repeated calls
- Greater trust in Geldium

Targets are pilot goals to be validated on Geldium data, not guaranteed results.